

India Air Quality Intelligence

Discovering Pollution Signatures, Temporal Patterns,
Station Variability & Data Quality

Data Analytics & Exploratory Data Analysis Project
CPCB CAAQMS Dataset | data.gov.in | Snapshot: 23 Sep 2026
Resource UUID: 3b01bcb8-0b14-4abf-b6f2-c1bfd384ba69

501

Monitoring Stations

263

Cities

31

States / UTs

84

Median Calc. AQI

71.7%

Full-Confidence AQI

71.5%

PM2.5 AQI Driver

IMPORTANT: All AQI values are CALCULATED AQI -- CPCB 2014 methodology. NOT official CPCB-reported AQI. Single cross-sectional snapshot (23 Sep 2026). No temporal trends can be established.

Technologies Used:

Python 3 (stdlib) | ECharts 5.4.3 | SVG | HTML5/CSS3 | ReportLab 5.x

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1. Executive Summary

This project presents a comprehensive exploratory data analysis of the Central Pollution Control Board (CPCB) Continuous Ambient Air Quality Monitoring Station (CAAQMS) dataset, sourced from the Government of India's Open Government Data Platform (data.gov.in). The dataset represents a **single cross-sectional snapshot** of 501 monitoring stations across 263 cities and 31 states/UTs of India, recorded at 23 September 2026, 12:00:00 IST.

All AQI values computed in this project are **Calculated AQI** — derived from the CPCB 2014 National AQI Framework using piecewise linear sub-index methodology. They are **not** the official CPCB-reported AQI values. Because this is a single snapshot, **no temporal trends, seasonal patterns, or changes over time can be established.**

Key findings at a glance:

#	Finding	Key Statistic
F1	PM2.5 dominates national AQI	71.5% of stations (358/501)
F2	Composition divergence at low AQI	10,700+ pairs: same AQI, different driver
F3	OZONE drives AQI in Odisha	52.6% of Odisha stations OZONE-driven
F4	High pollution = low diversity	Spearman $\rho = -0.619$
F5	Delhi within-city variability	AQI 41–388 across 45 stations simultaneously
F6	Unequal monitoring coverage	Top 5 states hold 55.5% of all stations
F7	NH3 never drives Calculated AQI	0 stations — breakpoints far above observed values
F8	Partial-confidence stations show lower AQI	Median 67.5 vs Full-confidence median 93.0

2. Problem Statement & Objectives

Air pollution is a pressing public health concern across India, with ambient air quality monitored at hundreds of stations by the CPCB CAAQMS network. However, raw monitoring data is rarely explored for its full analytical depth. Key questions remain unanswered:

- Which pollutants actually drive the Calculated AQI, and does this vary by region?
- Can two stations share similar AQI values yet have fundamentally different pollution profiles?
- Do cities exhibit characteristic pollution signatures distinct from the national average?
- How much does AQI vary *within* a single city, and what does this imply for city-level reporting?
- Is higher AQI always associated with a single dominant pollutant, or do multi-pollutant patterns exist?
- Where are monitoring data gaps most severe, and how does this affect analytical confidence?

Project Objectives

- Conduct rigorous data inspection, cleaning, and quality assessment on the raw CPCB CAAQMS dataset.
- Compute Calculated AQI for all 501 stations using the CPCB 2014 National AQI Framework.
- Identify dominant AQI-driving pollutants at national, state, city, and station levels.
- Detect and characterise compositional divergence — stations with similar AQI but different dominant pollutants.
- Quantify the relationship between AQI level and pollutant-profile diversity using Shannon entropy.
- Measure and visualise within-city AQI variability to expose limitations of city-level AQI averaging.
- Assess monitoring coverage inequality across states and identify data quality flags.
- Deliver a fully interactive 8-page analytical dashboard with documented methodology.

3. Dataset & Official Source

Source Details

Field	Value
Platform	Government of India Open Government Data Platform (data.gov.in)
Dataset Name	CPCB Real-time Air Quality Index Data
Publisher	Central Pollution Control Board (CPCB), MoEFCC, Government of India
Resource UUID	3b01bcb8-0b14-4abf-b6f2-c1bfd384ba69
Resource URL	https://data.gov.in/resource/3b01bcb8-0b14-4abf-b6f2-c1bfd384ba69
License	National Data Sharing and Accessibility Policy (NDSAP), Govt. of India
Snapshot Date/Time	23 September 2026, 12:00:00 IST
Data Type	Single point-in-time snapshot (NOT a time series)

Dataset Dimensions

Metric	Value
Total rows (raw)	3,507
Total columns (raw)	11
Monitoring stations	501 (each station appears exactly 7 times — once per pollutant)
Cities	263
States / Union Territories	31
Pollutants measured	7 (PM2.5, PM10, SO2, NO2, CO, NH3, OZONE)
NA readings	343 (9.8% of 3,507 records)
Date range	Single snapshot — 23 Sep 2026, 12:00:00 IST

4. Dataset Structure & Data Dictionary

The raw dataset is in **long format** — one row per station × pollutant combination. With 501 stations and 7 pollutants each, this yields 3,507 rows.

Raw Column Dictionary

Column	Type	Description	Use in Analysis
country	Categorical	Always "India"	Drop — no variance
state	Categorical	State/UT name	Geographic grouping
city	Categorical	City name	Geographic grouping
station	Identifier	Station name (includes board suffix after " - ")	Primary key
last_update	Datetime-like	Snapshot timestamp (dd-mm-yyyy hh:mm:ss)	Snapshot metadata only
latitude	Numeric	Station latitude (decimal degrees)	Map visualisation
longitude	Numeric	Station longitude (decimal degrees)	Map visualisation
pollutant_id	Categorical	Pollutant name (PM2.5, PM10, SO2, NO2, CO, NH3, OZONE)	AQI computation
pollutant_min	Numeric/NA	Minimum concentration in reporting period	Quality check
pollutant_max	Numeric/NA	Maximum concentration in reporting period	Quality check
pollutant_avg	Numeric/NA	Average concentration — used for AQI (343 NAs)	AQI computation

Important note on pollutant_avg: The averaging window (e.g., 24-hour, 8-hour) is **not documented** in the CSV. For AQI computation, it is **assumed** to be: 24-hour average for PM2.5, PM10, SO2, NO2, NH3; and 8-hour average for CO and OZONE — consistent with CPCB 2014 AQI methodology.

Key Engineered Columns (Wide Format — aqi_cleaned_wide.csv)

Column	Description
calculated_AQI	Composite AQI — max sub-index across available pollutants (CPCB 2014)
aqi_label	AQI category: Good / Satisfactory / Moderate / Poor / Very Poor / Severe
aqi_category	Confidence: FULL / PARTIAL_HIGH_CONFIDENCE / PARTIAL_LOW_CONFIDENCE / SINGLE_POLLUTANT_ONLY
dominant_pollutant_by_AQI	Pollutant with highest sub-index at this station
dominant_AQI_subindex	Value of the highest sub-index (determines Calculated AQI)
SI_PM2.5 ... SI_OZONE	Individual AQI sub-index values (0–500 scale) per pollutant
completeness_pct	Percentage of 7 pollutants with valid (non-null) readings
suspicious_zero_flag	TRUE if NH3 or SO2 = 0 at a fully-reporting station (12 stations)
possible_sensor_anomaly	TRUE if CO > 100 µg/m³ (flat reading pattern — 11 stations)
shannon_entropy	Shannon entropy of sub-index profile — pollutant diversity measure
{P}_zscore	Z-score per pollutant across all stations (for clustering/similarity only; NOT for AQI)

5. Data Cleaning & Quality

Cleaning Steps Applied

- **Numeric parsing:** pollutant_min, pollutant_max, pollutant_avg strings converted to floats; "NA" strings → None. New _num variant columns created; originals preserved.
- **Snapshot metadata:** last_update parsed to snapshot_date and snapshot_time. Column snapshot_label set to "SINGLE_SNAPSHOT_NOT_TIME_SERIES" — prevents misuse as time-series data.
- **CO unit conversion:** CO values are stored as $\mu\text{g}/\text{m}^3$ in the CSV. Divided by 1000 → mg/m^3 before applying CPCB AQI breakpoints. The $\mu\text{g}/\text{m}^3$ interpretation is confirmed by physical plausibility — treating them as mg/m^3 would classify every station as Severe, which is physically impossible.
- **Whitespace stripping:** 35 station names had leading/trailing whitespace — stripped.
- **Board acronym extraction:** extracted from station name suffix after " - " pattern.
- **No rows deleted:** all 3,507 rows retained. Suspicious zeros and anomalies are flagged, not removed.
- **No imputation:** NA values are propagated. AQI is computed from available sub-indexes only, with confidence category reflecting completeness.

Data Quality Summary

Quality Issue	Count	Impact	Action
Missing pollutant_avg (NAs)	343 (9.8%)	Reduces AQI confidence	Retained; AQI confidence category assigned
NH3 missingness	89 stations (17.8%)	No impact — NH3 never drives AQI	Retained; flagged
CO missingness	20 stations (4.0%)	Lowest-missingness pollutant	Retained
Suspicious zero — NH3	11 stations	NH3 sub-index = 1.0 (negligible)	Retained; suspicious_zero_flag=True
Suspicious zero — SO2	1 station	SO2 sub-index unaffected at these levels	Retained; suspicious_zero_flag=True
Possible sensor anomaly — CO	11 stations (CO > 100 $\mu\text{g}/\text{m}^3$)	CO may inflate Calculated AQI at these stations	Retained; possible_sensor_anomaly=True
High-missingness stations (≥ 3 NA)	41 stations	PARTIAL_LOW or SINGLE confidence AQI	Retained; confidence category reflects this
Duplicate station names	0	None	Confirmed clean
Out-of-bounds coordinates	0	None	All within India bounds (6–37°N, 68–98°E)

6. AQI Methodology

Calculated AQI is computed using the **CPCB 2014 National Air Quality Index framework** (piecewise linear sub-index formula). The composite AQI for each station is the **maximum sub-index** across all available pollutants. The pollutant with the highest sub-index is the dominant AQI driver.

Sub-index Formula

For a pollutant with observed concentration C , and breakpoints C_{lo} – C_{hi} mapping to sub-index range I_{lo} – I_{hi} :

$$SI = I_{lo} + (C - C_{lo}) \times (I_{hi} - I_{lo}) / (C_{hi} - C_{lo})$$

The composite Calculated AQI for a station is: $\max(SI_{PM2.5}, SI_{PM10}, SI_{SO2}, SI_{NO2}, SI_{CO}, SI_{NH3}, SI_{OZONE})$

CPCB 2014 AQI Breakpoints Reference

Pollutant	Avg Period	Unit	Good (0–50)	Satisfactory (51–100)	Moderate (101–200)	Poor (201–300)	Very Poor (301–400)	Severe (401–500)
PM2.5	24h	µg/m³	0–30	30–60	60–90	90–120	120–250	250–380
PM10	24h	µg/m³	0–50	50–100	100–250	250–350	350–430	430–600
SO2	24h	µg/m³	0–40	40–80	80–380	380–800	800–1600	1600–2100
NO2	24h	µg/m³	0–40	40–80	80–180	180–280	280–400	400–800
CO	8h	mg/m³*	0–1	1–2	2–10	10–17	17–34	34–46
NH3	24h	µg/m³	0–200	200–400	400–800	800–1200	1200–1800	1800–2400
OZONE	8h	µg/m³	0–50	50–100	100–168	168–208	208–748	748–1000

* CO values in the dataset are in µg/m³. Divided by 1000 to convert to mg/m³ before applying breakpoints. The 8h/24h averaging periods are assumed (not documented in CSV).

AQI Confidence Categories

Category	Label	Criteria	Count	Median Calc. AQI
FULL	A	All 7 pollutant sub-indexes available	359	93.0
PARTIAL_HIGH_CONFIDENCE	B	4–6 sub-indexes valid	110	67.5
PARTIAL_LOW_CONFIDENCE	C	1–3 sub-indexes valid	26	67.5*
SINGLE_POLLUTANT_ONLY	D	Only 1 sub-index available	6	N/A

* B and C grouped together as "Partial-confidence" for Finding F8.

7. Methodology Assumptions & Limitations

Averaging period [ASSUMED — not documented in CSV]

The pollutant_avg column's averaging window is not specified. Assumed: 24h for PM2.5/PM10/SO2/NO2/NH3; 8h for CO/OZONE (per CPCB 2014 standard). Actual averaging period may differ, which would affect sub-index magnitudes.

CO unit [$\mu\text{g}/\text{m}^3$ — confirmed by physical plausibility]

CO values in the dataset (range 1–196) are in $\mu\text{g}/\text{m}^3$, not mg/m^3 . The mg/m^3 interpretation would classify 100% of stations as Severe — physically impossible. Divided by 1000 before applying CPCB breakpoints.

Single snapshot [Point-in-time only]

The entire dataset represents a single moment. No temporal trends, seasonality, diurnal patterns, or changes over time can be established or inferred.

Calculated $\neq$ Official AQI [Recomputed, not taken from source]

All AQI values are independently computed using CPCB 2014 methodology. The source file contains a "pollutant_avg" field — not an official AQI field. Results may differ from CPCB's own published AQI values.

No imputation [NA values retained as-is]

Missing pollutant readings are not imputed. The composite AQI is computed from available sub-indexes only. Stations with fewer available pollutants receive a lower confidence category.

No causal inference [Observational only]

All associations in this project are cross-sectional and observational. No causal claims are made about pollution sources, health outcomes, or industrial activity.

Coverage inequality [Unequal station distribution]

Top 5 states hold 55.5% of all monitoring stations. State-level AQI comparisons are not directly comparable — states with fewer stations have higher uncertainty in any aggregate statistic.

8. EDA Methodology

The following analytical techniques are used in this project. All computations are performed in pure Python (standard library) without external statistical packages.

AQI Sub-index Computation: Piecewise linear interpolation per CPCB 2014. One sub-index per pollutant per station; composite = max sub-index.

Dominant Pollutant Identification: Pollutant with the highest sub-index per station. Aggregated at city, state, and national levels.

Shannon Entropy (Pollutant Diversity): $H = -\sum p_i \times \log(p_i)$ where p_i is the proportion of each pollutant's sub-index in the total sub-index sum. Higher entropy = more balanced multi-pollutant profile. Range: 0 (single driver) to $\log(7) \approx 1.95$.

Spearman Rank Correlation: Non-parametric rank correlation between Calculated AQI and Shannon entropy across 495 stations with valid values. Measured association (not causation).

Normalized Pollutant Profiles: City median concentration $\div$ national 75th percentile. Values > 1.0 indicate above the national 75th percentile for that pollutant. Used for city "pollution signature" heatmap.

Within-city Variability: For cities with ≥ 2 stations: range, std dev, IQR, and coefficient of variation of Calculated AQI across constituent stations.

AQI Band Composition: Stations grouped into 6 AQI bands (0–50, 51–100, etc.). Stacked-bar showing dominant pollutant distribution per band reveals composition divergence.

Quadrant Analysis: Stations classified into 4 quadrants using AQI P75 (190) and entropy median (1.30) as thresholds: High/Low AQI $\times$ High/Low Diversity.

9. Key Findings

Finding F1: PM2.5 Dominance

F 1 PM2.5 drives Calculated AQI at 71.5% of stations (358/501)

PM2.5 is the dominant AQI-driving pollutant at 358 of 501 monitoring stations nationally. Above AQI 150, PM2.5 drives 100% of stations — compositional diversity in AQI drivers is exclusively a low-to-moderate pollution phenomenon.

Risk: PM2.5-focused interventions may miss stations where other pollutants drive AQI at lower levels.

Opportunity: PM2.5 reduction yields the highest AQI improvement per intervention unit at the most-polluted stations.

Action: Prioritise PM2.5 source apportionment at high-AQI stations. At AQI < 150, assess station-specific driver before intervention design.

Finding F2: Composition Divergence

F 2 10,700+ station-pairs with same AQI (± 20) but different dominant pollutants

Stations with nearly identical Calculated AQI values can be driven by entirely different pollutants. This divergence is entirely confined to AQI below 150. At AQI > 150, all stations are PM2.5-driven. This means a single AQI number is insufficient to characterise the pollution profile at moderate and lower levels.

Risk: City-level single-AQI reporting at AQI < 150 may direct wrong-pollutant interventions.

Opportunity: Station-level sub-index profiles enable source-specific pollution management even at moderate AQI.

Action: Report dominant pollutant alongside AQI category. Do not use AQI alone to select interventions at AQI < 150.

Finding F3: OZONE Regional Signature (Odisha)

F3 Odisha is the only state where OZONE is the plurality AQI driver (52.6% of stations)

OZONE drives Calculated AQI at 45 stations nationally (9%). However, the geographic distribution is highly uneven — Odisha accounts for a disproportionate share, with 52.6% of its monitoring stations being OZONE-driven. This is a distinct regional pollution signature not visible in national-level aggregates.

Risk: National PM2.5-focused air quality policies may miss OZONE-dominant regions like Odisha.

Opportunity: Odisha presents a natural "laboratory" for OZONE reduction strategies separate from PM2.5 interventions.

Action: Review Odisha monitoring data for photochemical precursor emissions (NOx, VOCs). Tailor state-level action plan.

Finding F4: Concentration vs Pollutant Diversity

F4 Spearman ρ (Calculated AQI, Shannon Entropy) = -0.619

A strong negative association exists between a station's Calculated AQI and the diversity of its pollutant sub-index profile. High-pollution stations are almost exclusively PM2.5-dominated (single-pollutant). Low-pollution stations show more balanced multi-pollutant profiles. Association only — no causation is implied.

Risk: Severe-category stations are analytically simpler (single driver) but represent the highest health risk.

Opportunity: 9 stations in the High-AQI / High-Diversity quadrant suggest co-elevated PM2.5+SO2 — possible industrial zones.

Action: At high AQI, PM2.5 mitigation is sufficient. The 9 high-AQI/high-diversity stations merit multi-pollutant source audit.

Finding F5: Within-City Variability (Delhi)

F5 Delhi: AQI 41–388 across 45 stations simultaneously (range = 347, std = 82.9)

Delhi's 45 monitoring stations span from AQI 41 (Good) to AQI 388 (Very Poor) in a single snapshot. A person at the "best" Delhi station experiences Good air; a person 10 km away experiences Very Poor air — simultaneously. City-level AQI averages conceal this heterogeneity, which has direct implications for exposure assessment.

Risk: City-average AQI misleads residents and policy-makers about their actual exposure.	Opportunity: Sub-city AQI mapping enables targeted interventions at the neighbourhood level.
Action: Report AQI distribution (box plot, range) for multi-station cities, not just the mean. Identify high-AQI pockets within cities.	

Finding F6: Monitoring Coverage Inequality

F6 Top 5 states hold 55.5% of all stations; 9 states/UTs have only 1 station

Monitoring infrastructure is highly concentrated in a few states. The top 5 states (Delhi, Uttar Pradesh, Maharashtra, West Bengal, Karnataka) account for 278 of 501 stations. Nine states and Union Territories have only a single monitoring station, making state-level AQI estimates from these regions statistically unreliable.

Risk: Low-coverage states may have unreported pollution hotspots. Single-station states have very high uncertainty.	Opportunity: Expanding monitoring to under-represented states and regions improves national AQI policy evidence base.
Action: Flag single-station states in any reporting. Prioritise monitoring expansion in North-East, Himalayan, and coastal UT regions.	

Finding F7: NH3 Never Drives Calculated AQI

F7 NH3 drives Calculated AQI at 0 of 501 stations

Despite being monitored at 412 of 501 stations, NH3 has the highest missingness rate (17.8%). More critically, all observed NH3 values (maximum = 40 µg/m³) fall far below the CPCB "Good" threshold of 200 µg/m³. NH3 sub-indexes are therefore always < 10, and NH3 never determines the composite AQI at any station.

Risk: NH3 monitoring resources are not contributing to AQI determination at current observed concentrations.

Opportunity: NH3 data could support independent ammonia-specific analysis (agriculture, waste) beyond the AQI framework.

Action: NH3 missingness (17.8%) is analytically inconsequential for AQI computation but should be reported in monitoring reviews.

Finding F8: Confidence–AQI Relationship

F8 Full-confidence stations: median AQI 93.0 vs Partial-confidence: 67.5

Stations with all 7 pollutant sub-indexes available (Full-confidence) have a systematically higher median Calculated AQI (93.0) than partial-confidence stations (67.5). This suggests that missing sensor data is not random — stations with fewer sensors tend to be in lower-pollution areas, or sensors are selectively offline.

Risk: Missing sensors may be systematically correlated with location type, understating pollution at partially-monitored stations.

Opportunity: Calibrating national AQI statistics to account for confidence-level distribution would improve representativeness.

Action: Report confidence distribution alongside national AQI statistics. Prioritise full-sensor deployment at urban core stations.

10. Fact → Insight → Risk → Opportunity → Action

Finding	Fact	Insight	Risk	Opportunity	Action
F1 PM2.5	71.5% of stations PM2.5-driven; 100% above AQI 150	PM2.5 is the dominant national AQI driver, especially at high-pollution sites	Complacency toward other pollutants at moderate AQI	PM2.5 reduction yields highest AQI gain per intervention	Station-specific driver assessment for AQI < 150; PM2.5 priority at AQI > 150
F2 Divergence	10,700+ pairs: same AQI, different dominant pollutant	Single AQI number is insufficient to characterise pollution profile below AQI 150	Wrong-pollutant interventions from AQI-only reporting	Sub-index profiles enable source-specific management	Report dominant pollutant alongside every AQI value
F3 OZONE	Odisha: 52.6% of stations OZONE-driven (national = 9%)	Regional pollution signatures differ sharply from national averages	National PM2.5 policy misses OZONE-dominant regions	Region-specific intervention design (precursor emissions)	Tailored state AQI action plans; investigate NOx/VOC sources in Odisha
F4 Diversity	Spearman ρ = -0.619 (AQI vs entropy)	Severe pollution is compositionally simple; moderate is diverse	Multi-pollutant complexity at low AQI is analytically underappreciated	9 high-AQI/high-diversity stations warrant industrial source audit	Multi-pollutant audit at High-AQI/High-Diversity stations
F5 Variability	Delhi AQI range = 347 in single snapshot (n=45 stations)	City-average AQI masks neighbourhood-level exposure disparities	City-level reporting misleads residents and policy-makers	Sub-city pollution mapping for targeted interventions	Publish AQI distribution (not just mean) for multi-station cities
F6 Coverage	9 states/UTs have 1 station; top 5 states = 55.5% of network	National AQI picture is biased toward well-monitored states	Underrepresented regions may have unreported pollution hotspots	Infrastructure expansion improves national evidence base	Monitoring expansion plan for North-East, Himalayan, island UTs

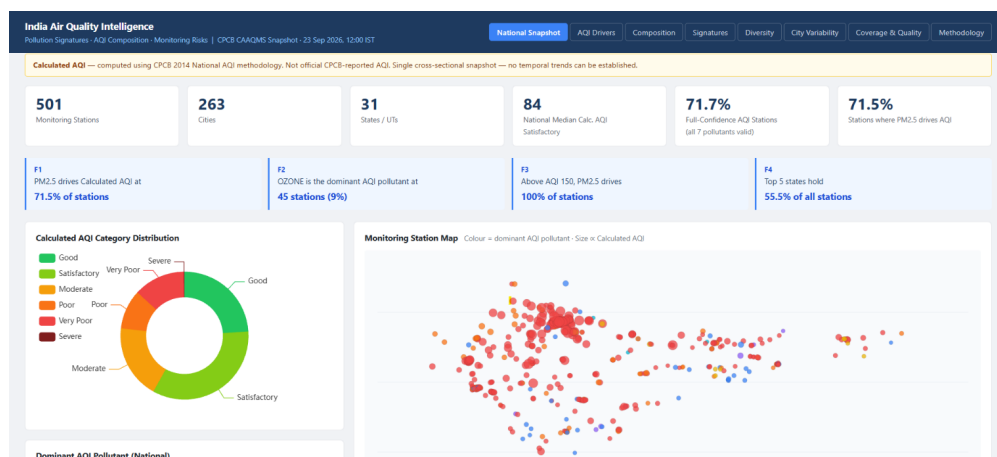
11. Dashboard Overview

The project deliverable is an interactive 8-page HTML dashboard (**dashboard.html**), built with ECharts 5.4.3 (CDN) and an SVG-based India station map. It loads pre-processed data from **dashboard_data.json** at runtime. No server-side processing is required — serve with `python -m http.server 8000` and open `http://localhost:8000/dashboard.html`.

Page	Title	Charts / Components	Key Analytical Question
1	National Snapshot	AQI category donut, dominant pollutant bar, SVG station map (501 dots)	What is the national AQI distribution?
2	AQI Drivers	Driver bar, state stacked bar, sub-index profile, station table (filterable by state/city)	Which pollutants drive Calculated AQI?
3	Composition Divergence	Band stacked bar, distinct-drivers bar, station sub-index stacks by AQI band	Same AQI, different composition?
4	Pollution Signatures	City normalised heatmap (40 cities × 7 pollutants), radar chart per city	Do cities have distinct pollution fingerprints?
5	Concentration vs Diversity	Entropy scatter plot (495 points, 3 colour modes), quadrant bar chart	Does higher AQI mean lower pollutant diversity?
6	Within-City Variability	Station AQI bar, sub-index radar overlay, city range bar (top 20 cities)	How much does AQI vary within a city?
7	Coverage & Data Quality	State stations bar, AQI confidence bar, pollutant NA bar, flag tables	Where are data gaps most severe?
8	Methodology	CPCB breakpoint table, methodology grid, disclaimer	How was Calculated AQI computed?

Dashboard Design Principles

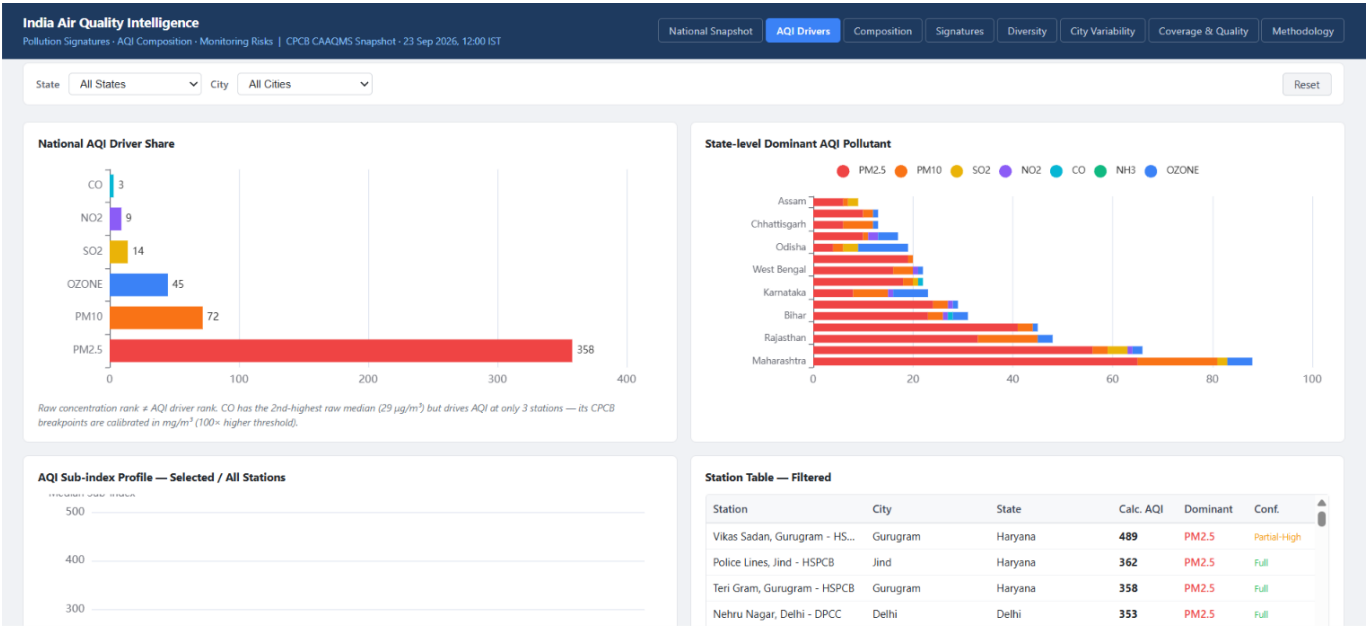
- Every AQI value displayed alongside: number of stations, dominant pollutant, AQI confidence category.
- City AQI shown as distribution (range / std dev / IQR), not a single point estimate.
- All charts carry a disclaimer identifying values as "Calculated AQI" (not official CPCB AQI).
- Station is the primary analytical unit — no city or state averages presented without caveat.
- No decorative charts — every visualisation answers a specific analytical question.
- Cross-sectional limitation stated on every page that could be misread as temporal.



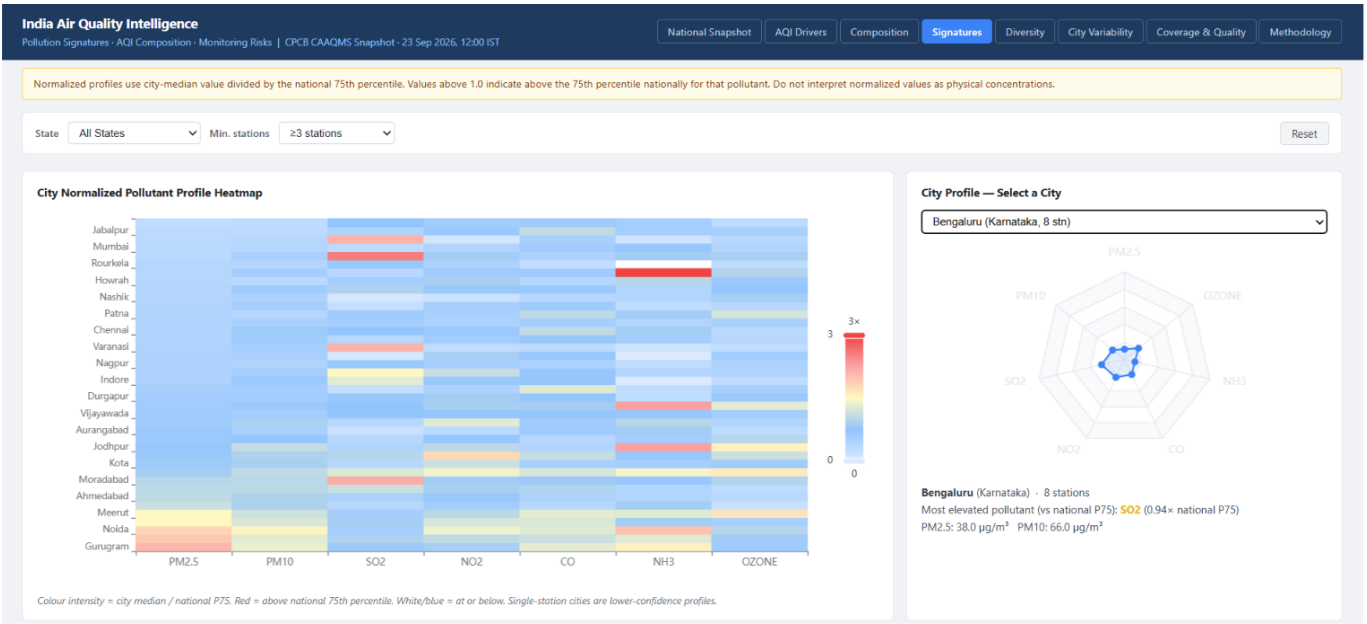
Dashboard Screenshot — National Snapshot

11. Dashboard Screenshots

Interactive dashboard views — India Air Quality Intelligence



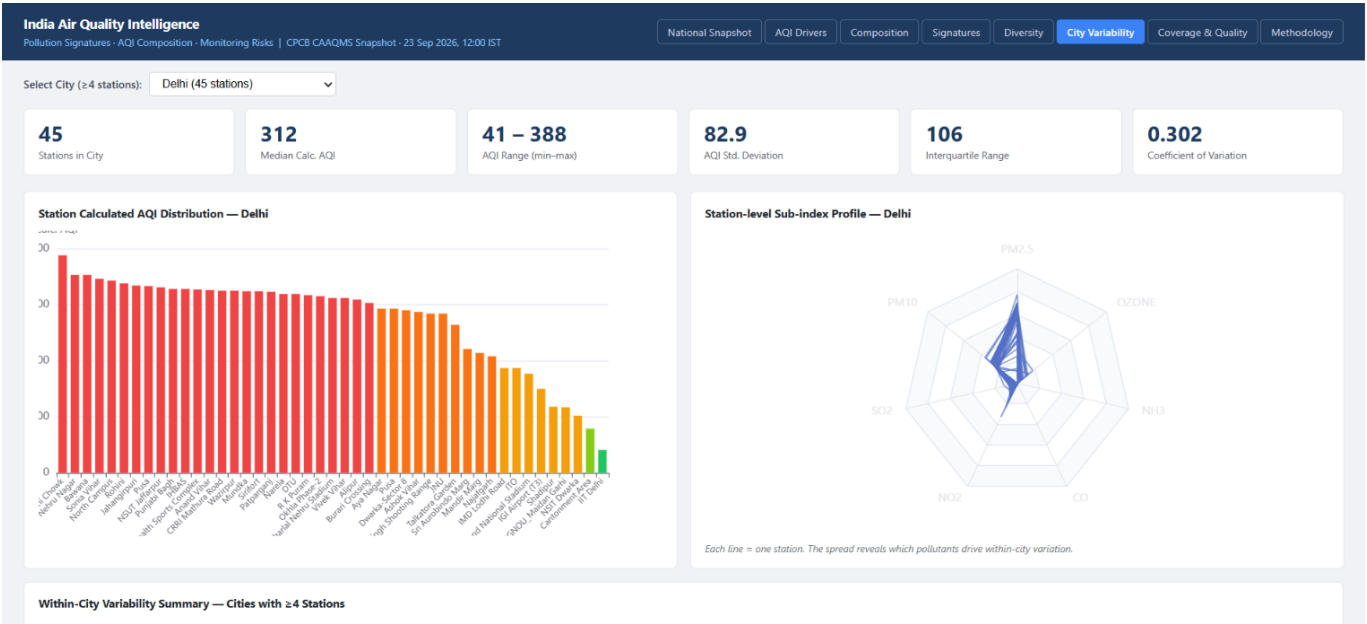
Dashboard Screenshot — AQI Drivers



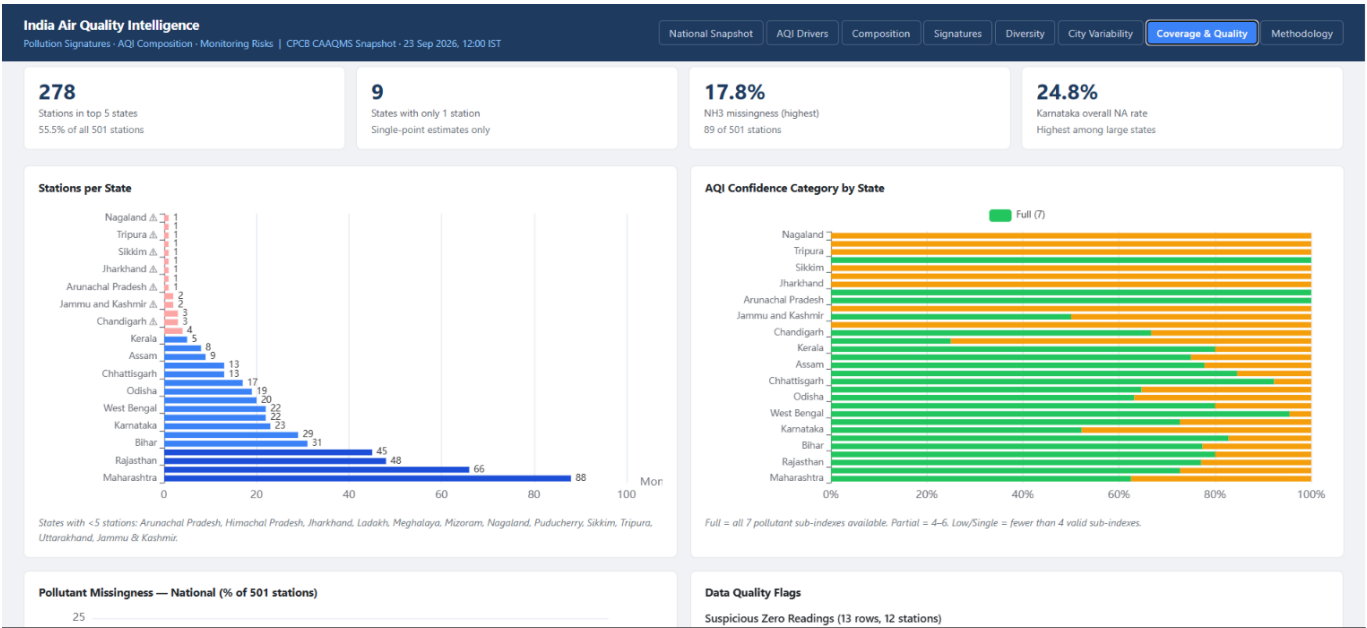
Dashboard Screenshot — Pollution Signatures

11. Dashboard Screenshots

Interactive dashboard views — India Air Quality Intelligence



Dashboard Screenshot — Within-City Variability



Dashboard Screenshot — Coverage & Data Quality

12. Analytical Limitations

Single cross-sectional snapshot: The dataset is a single point-in-time recording. No temporal trends, seasonal variation, diurnal patterns, or changes over time can be established.

Averaging period undocumented: The pollutant_avg field's averaging window is not documented in the CSV. Assumed to follow CPCB 2014 (24h for PM/SO₂/NO₂/NH₃; 8h for CO/OZONE). If the actual averaging window differs, sub-index values may be incorrect.

Calculated ≠ Official AQI: All AQI values are recomputed, not taken from an official CPCB source. Results may differ from CPCB's published AQI for the same stations.

CO sensor anomalies: 11 stations show CO > 100 µg/m³ (flat readings). Calculated AQI may be inflated at these stations. Results are retained with a flag.

Suspicious zeros: 12 stations show NH₃ or SO₂ = 0 at fully-reporting stations. These may be sensor calibration artefacts. Retained and flagged.

Coverage inequality: Top 5 states account for 55.5% of stations. AQI comparisons across states are not directly comparable without accounting for this imbalance.

Observational only — no causal inference: All associations (e.g., PM_{2.5} dominance, entropy–AQI correlation) are observational. No causal claims about sources, health outcomes, or industrial activity are made or supported by this data.

Missing geographic context: No land-use, traffic density, industrial proximity, or meteorological data is available. Pollution signatures cannot be attributed to specific sources.

No health outcome data: AQI categories describe air quality only. No health impact quantification is performed or implied.

13. Conclusion

This project demonstrates that a single cross-sectional snapshot of CPCB CAAQMS data contains substantially more analytical depth than is revealed by city-level AQI averages. The eight key findings — from PM_{2.5}'s near-universal dominance at high pollution levels, to the striking within-city variability in Delhi, to OZONE's regional concentration in Odisha — each challenge simplistic national-level air quality narratives.

The composition divergence finding (10,700+ station-pairs with similar AQI but different dominant pollutants) highlights that single-number AQI reporting can actively mislead intervention design at moderate and lower pollution levels. The entropy analysis (Spearman $\rho = -0.619$) provides a quantitative basis for this observation.

The monitoring coverage analysis reveals that 9 states and UTs have only one monitoring station, and the top 5 states dominate the network. This structural imbalance must be acknowledged in any national air quality policy formulation.

All analytical conclusions are qualified by the single-snapshot limitation — no temporal inference is possible. The project methodology is fully documented, reproducible via `main.py`, and does not make causal claims.

14. Future Scope

- **Time-series analysis:** With multiple snapshots or the full CPCB hourly data, temporal trends, seasonal patterns, and diurnal AQI cycles could be investigated.
- **Source apportionment:** Integration of land-use, industrial zone, and traffic density data could support observational associations between pollution profiles and source types.
- **Meteorological correlation:** Combining AQI data with wind, humidity, and temperature data from IMD would enable weather-AQI relationship analysis.
- **Health outcome linkage:** Integration with hospital admission or respiratory disease data (with appropriate caution regarding causation) could quantify population exposure.
- **Monitoring gap prioritisation:** A formal network-optimisation analysis could identify which unmonitored districts would most benefit from new CAAQMS installations.
- **Clustering/classification:** Station-level pollution profiles (z-scores computed in main.py) could be used for unsupervised clustering to identify station typologies.
- **AQI forecast model:** With time-series data, machine learning models (LSTM, XGBoost) could be trained to forecast Calculated AQI 24–48 hours ahead.

15. References & Data Source

Primary Data Source

- Government of India Open Government Data Platform (data.gov.in)
- Dataset: CPCB Real-time Air Quality Index Data
- Publisher: Central Pollution Control Board (CPCB), Ministry of Environment, Forest and Climate Change
- Resource UUID: 3b01bcb8-0b14-4abf-b6f2-c1bfd384ba69
- URL: <https://data.gov.in/resource/3b01bcb8-0b14-4abf-b6f2-c1bfd384ba69>
- License: National Data Sharing and Accessibility Policy (NDSAP), Government of India

AQI Methodology

- Central Pollution Control Board (2014). National Air Quality Index.
- Ministry of Environment, Forest and Climate Change, Government of India.
- Document defines the piecewise linear sub-index formula and breakpoints used in this project.

Software & Libraries

- Python 3 — standard library (csv, json, math, os, sys, collections, statistics)
- ECharts 5.4.3 — Apache ECharts (<https://echarts.apache.org/>) — CDN: jsdelivr.net
- ReportLab 5.x — PDF generation (<https://www.reportlab.com/>)
- Dashboard: HTML5 / CSS3 / JavaScript (vanilla, no framework)

IMPORTANT DISCLAIMER: All AQI values in this report are **Calculated AQI** — independently computed using the CPCB 2014 National AQI Framework. They are **not** official CPCB-reported AQI values. This dataset represents a single cross-sectional snapshot (23 September 2026, 12:00:00 IST). **No temporal trends, seasonality, or changes over time are established or implied.** All associations are observational. No causal claims about pollution sources or health outcomes are made.